CyberShield AI: An Explainable Multi-Model Artificial Intelligence Framework for Integrated Cyber Threat Detection

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***Abstract*— Rapid growth of communication, online payments, social media and multimedia platforms have increased users' exposure to a wide range of cyber threats. In addition to attacks cyber threats such as phishing, malicious URLs, fraudulent QR codes, manipulated images and videos and synthetic audio are becoming increasingly difficult to detect as they often imitate legitimate digital content. Artificial Intelligence, Machine Learning and Deep Learning have therefore become research areas for automated cyber threat detection. However existing solutions are being developed for individual cyber threat categories. Differ greatly in their input types, detection methods, computational demands, adaptability and level of explainability.**

***This review paper examines AI-based approaches for detecting and addressing multiple categories of cyber threats, including phishing, malicious URLs, QR-code and payment-related threats, image and video deepfakes and audio deepfakes. The review also considers the role of Explainable Artificial Intelligence in making security decisions easier for users and analysts to understand. Different techniques, including Natural Language Processing, transformer-based models, Convolutional Neural Networks, spectral audio representations, computer vision, QR-code analysis and hybrid Machine Learning approaches are. Compared.**

***The review identifies common challenges in existing literature, including the separation of threat-specific detection systems difficulty in handling evolving and previously unseen attacks, computational overhead, limited interpretability and insufficient emphasis on actions that can be taken after a cyber threat is detected. These observations highlight the need to move from systems toward security applications that can assess risk explain decisions and support preventive action.**

***Based on these findings the paper discusses the motivation and design perspective of CyberShield AI as an application-oriented approach that brings multiple threat analysis capabilities within a unified detection, risk assessment, explanation and prevention workflow.**

***The paper concludes by outlining opportunities, for developing more adaptive explainable, efficient and user-oriented AI-based cybersecurity applications.**

***Keywords*— Artificial Intelligence, Cybersecurity, Machine Learning, Deep Learning, Multi-Modal Threat Detection, Phishing Detection, Malicious URL Detection, QR Code Security, Deepfake Detection, Explainable Artificial Intelligence, Cyber Threat Prevention**

I INTRODUCTION

Cybersecurity has become a critical global concern as online communication, financial services, cloud platforms, social media, and Internet-connected devices have become a part of everyday life. As a result, attackers have found numerous ways to leverage technical vulnerabilities and user behavior. Phishing, malicious websites, fraudulent digital-payment requests, QR-code-based attacks, manipulated images, and synthetic media are increasingly used in various forms and are sometimes combined within the same attack campaign. Traditional security mechanisms that rely on signatures, rules, and blacklists have proven to be useful in identifying known threats. However, their effectiveness is significantly limited by frequent modification of content or the introduction of new, previously unobserved attack patterns, [9].

As a result, Artificial Intelligence (AI) and Machine Learning (ML) have become the focus of research in cybersecurity. Researchers have explored conventional machine learning algorithms such as Random Forest, Support Vector Machine (SVM), Decision Tree, Naïve Bayes, and XGBoost for cybersecurity data classification. On the other hand, Deep Learning (DL) models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), and transformer-based architectures have been increasingly used for textual, sequential, and visual threat detection [2]–[7]. For example, CNN approaches for URL detection have indicated that character-level representations can be effective for classification without relying on manually defined URL rules. Similarly, phishing detection research has investigated both machine learning techniques and Natural Language Processing (NLP) methods for identifying suspicious email content alongside related contextual information, [3].

The literature, however, shows that superior classification performance does not necessarily offer holistic cybersecurity protection. Phishing detection systems commonly examine message content, sender information, headers, embedded URLS, and other textual characteristics, while malicious URL detection systems mainly focus on host-based or behavioral properties of websites, [4]. QR-code security studies focus on decoding embedded links and checking their

reputation, while image and deepfake detection approaches generally examine artifacts, spatial inconsistencies, and facial features in an isolated manner [5]–[8]. As a result, most existing solutions are designed for a single threat category versus being able to combine different types of evidence from attack surfaces.

The separation becomes more dangerous with the idea that modern cyberattacks can involve more than one type of digital content. For example, a phishing email may involve a malicious URL or QR code, while a fraudulent QR-code may direct a victim to a malicious payment page. Similarly, manipulated screenshots can be used to make a financial scam appear convincing, and synthetic images, videos, or voices can be used for sophisticated social-engineering attacks. Research on QR-code security has indicated the close connection between QR-codes and malicious URLs, while recent work on fraud has identified the growing use of machine learning for detecting unusual transaction behavior, [10]. These examples suggest that examining each artifact separately does not provide a complete picture of the associated security risk.

Another major problem is the changing nature of AI-assisted attacks. Generative AI can now be used to mass-produce highly convincing text, synthetic images, manipulated videos, and artificial voices. As such, it is becoming harder for both users and automated systems to rely on the linguistic clues that have traditionally been used to identify fraudulent content. Recent deepfake detection studies have reported that detection models become significantly less reliable when they encounter new generation techniques, compressed media, and cross-dataset testing environments, [8]. Similarly, research on URL detection has identified adversarial manipulation and changing URL structures as critical roadblocks for machine learning-based detectors, [12].

The dependence on datasets is another persistent issue that has been observed across the literature. Many proposed approaches are evaluated using fixed datasets with static attack distributions. While such datasets have been important for controlled evaluation, they do not fully represent the dynamic nature of real-world attacks. In URL detection, for example, attackers can instantly register new domains, change URL tokens, and deploy unique obfuscation techniques. Research on adversarial attacks against URL detectors has shown that carefully modifying specific URL tokens can easily subvert machine learning predictions. These findings have raised questions regarding how effective a model trained on a particular dataset would perform when it encounters new or dynamically modified attack examples.

Alongside raw classification performance, explainability has come to be an essential requirement in AI-based cybersecurity. Complex deep learning models can produce a high-accuracy classification result without showing which underlying characteristics influenced that decision. As a result, it is exceptionally hard for cybersecurity analysts to understand, validate, and trust alerts before taking critical defensive action. Explainable Artificial Intelligence (XAI) techniques such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) can provide local feature attributions to make AI decisions easier to interpret. The literature, however, suggests that explainability is often studied in isolation and is not integrated into the challenge of detecting threats across multiple domains simultaneously.

With these observations, a comprehensive review that looks at classification accuracy beyond the domain of individual models is the motivation for this paper. The purpose of this paper is to examine AI-based cybersecurity research across multiple threat domains and identify common pitfalls in methodology and practice. This review focuses specifically on phishing email detection, malicious URL classification, QR-code-based threats, image manipulation, deepfake detection, and related financial fraud-detection research. The existing approaches are critically examined in terms of their algorithms, input characteristics, detection strategies, benchmark datasets, reported performance, and constraints. By directly comparing these areas, this review aims to establish how current AI-based cybersecurity approaches can move away from isolated threat detection and transition into integrated and prevention-oriented security applications.

The major contributions of this review are organized as follows:

1. Cross-Domain Systematic Synthesis: A cross-domain evaluation of intelligence-based methods for detecting phishing attacks, malicious URLs, QR-code threats, image manipulation, deepfake, and concurrent digital fraud.

2. Technical Taxonomy Comparison: A detailed comparison of the technical architectures utilized in the literature, including traditional machine learning, deep learning, natural language processing, computer vision, and transformer-based models.

3. Core Research Gap Identification: Systematic identification of persistent research vulnerabilities across studies, such as lack of cross-domain integration, heavy reliance on static datasets, poor zero-day defense, weakness against adversarial manipulation, lack of model explainability, and real-time deployment constraints.

4. XAI Integration Analysis: An evaluation of transparency and trust requirements in modern cybersecurity systems, especially in how techniques such as SHAP and LIME can be integrated to support human security analysts.

5. Unified Reference Architecture: A proposed conceptual framework, CyberShield AI, synthesized from identified literature shortcomings, intended to guide future research towards unified, multimodal, and explainable cyber threat assessment pipelines.

CyberShield AI is presented in this review as a conceptual framework and future research direction instead of an active, finalized software system. The overarching objective is to bridge the gaps found across existing independent studies, which lays out the foundations of building integrated defensive applications that are highly adaptive and user-oriented.

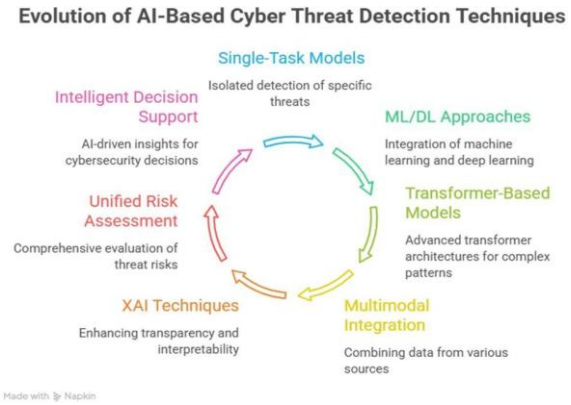


Figure 1. Evolution of AI-based Cyber Threat Detection Techniques from Single-Task Models to Integrated Explainable Multimodal Frameworks.

II. RESEARCH METHODOLOGY

To ensure transparency to the article structure, prevent selection bias, and allow absolute reproduction of our findings, we applied a structured, multi-step review methodology to identify and synthesize the relevant literature.

A. DATABASE SOURCES AND SEARCH STRATEGY

We leveraged five major indexing databases to perform primary literature searches: IEEE Xplore, ACM Digital Library, Scopus, ScienceDirect, and Google Scholar. We imposed a strict publication date range between January 2021 and October 2026 to capture the latest advances in generative artificial intelligence, transformer-based architectures, and explainable cybersecurity approaches.

The following database search strings were constructed using Boolean operators (AND/OR) and combined with unique database threat signatures/interpretability tokens:

("Artificial Intelligence" OR "Deep Learning" OR "Machine Learning")

AND ("Phishing Detection" OR "Malicious URL" OR "QR Code Security" OR "Deepfake Video" OR "Image Manipulation Forensic")

AND ("Explainable AI" OR "XAI" OR "SHAP" OR "LIME")

B. INCLUSION/EXCLUSION CRITERIA

The inclusion/exclusion criteria were applied to the initial search results to identify relevant studies for synthesis:

Inclusion Criteria: (1) Peer-review journal articles, transaction papers, or core international conferences; (2) Papers reporting algorithmic models/design specifications or quantitative experimentation/validation; (3) Contributions analyzing out-of-distribution generalization, multi-modal data fusion, or active interpretability.

Exclusion Criteria: (1) Non-peer-review preprints, industry whitepapers, patents, or extended abstracts; (2) Works focusing exclusively on network-layer intrusion detection systems (IDS) or hardware centric security; (3) Frameworks missing baseline evaluation metrics or performance validation sets.

C. SCREENING AND DATA EXTRACTION

A multi-stage screening process was applied to the initial database results. First, we removed duplicate records. Second, we applied title/abstract screening to identify relevant works. Finally, we performed a full-text review to ensure inclusion/exclusion criterion compliance and narrowed our synthesis corpus to a set of unique, high-quality papers.

For each included work, we extracted the following information in a standardized field matrix to establish our baseline of verifiable contributions:

Benchmark Datasets: (e.g., ISCX-URL2016, FaceForensics++, Celeb-DF) [22]

Algorithmic Modalities: (Text, lexical strings, visual pixel grids, temporal frames) [8]

Peak Performance Metrics: (Accuracy, AUROC margins)

Deployment Settings: (Lab-bound controlled setups vs. live production/browser runtime environments) [12]

Explicit Limitations: (Vulnerabilities, adversarial blindspots, processing latency)

III LITERATURE REVIEW

Artificial Intelligence and Machine Learning have emerged as essential technologies for cyber security as they allow for automated discovery and analysis of suspicious digital objects. Contemporary cyber threats have evolved beyond traditional phishing schemes, and researchers are now looking to detect malicious URLs, QR code content, forged images and deepfake videos, and various social engineering techniques. Traditional signature-based, rule-based, and blacklist approaches are only partially effective as they are

only as good as the previously discovered attacks allowing attackers to circumvent them by modifying known malicious patterns.

Various research works explored the application of conventional machine learning, deep learning, natural language processing, and image analysis to detect threats, each targeting a particular class of attacks. While individual solutions often demonstrate high accuracy for a particular use case, the research area lacks a holistic approach that would consider the entire attack chain of multi-stage phishing campaigns. This section provides a brief overview of solutions addressing email phishing detection, malicious URL discovery, QR code content analysis, image forgery detection, and deepfake identification. For each area, the discussion highlights the core algorithms, input data, detection mechanisms, advantages, and limitations.

A. Email Phishing Detection

Email phishing has become one of the most pervasive cyber threats as it often relies on the persuasion of the victim to follow the attacker's instructions. Attackers often utilize social engineering techniques and malicious attachments or websites to gain unauthorized access to the target's system and data. Traditional spam filters only offer limited protection as they are based on simple keyword matching, sender reputation assessment, and blacklisted domains and addresses. These techniques are relatively easy to circumvent, especially when advanced persistent threats launch targeted spear-phishing campaigns or utilize large language models to generate unique messages for each spam campaign [6], [7].

Researchers employed both traditional machine learning and deep learning techniques to detect phishing emails and messages with better accuracy. Various works demonstrated that classical machine learning algorithms such as Random Forest, SVM, Naïve Bayes, Decision Trees, or XGBoost combined with handcrafted feature extraction from email headers, routing traces, embedded URLs, attachments, HTML code, and other data demonstrate relatively high accuracy and are computationally efficient [6], [8]. On the other hand, deep learning solutions attempt to eliminate the need for feature engineering by utilizing raw data as the input. Stacked recurrent layers combined with CNN can be used to learn sequential patterns embedded in email messages while the transformer architecture demonstrates exceptional effectiveness for processing natural language [7], [9]. Despite the overall high accuracy reported by many works, most solutions fail to assess performance on the latest zero-day campaigns, multilingual spam, and advanced spear-phishing messages. Moreover, most solutions suffer from the lack of explainability as complex deep learning models are not easily interpretable by security analysts, who may want to investigate the suspicious email [6], [7], [10].

B. Malicious URL Detection

Malicious URLs are often the primary carriers of phishing campaigns, malware distribution, credential theft, and financial fraud schemes. Traditional detection mechanisms are mostly based on regular expressions and are effective for detecting known malicious domains, shortened URLs, phishing kits, and other known resources. They are limited when novel domains with suspicious character patterns or newly established phishing sites are involved [11], [12]. Various research works demonstrated that machine learning is an effective way to complement traditional string matching and enhance the robustness of phishing URL detection mechanisms. Popular algorithms such as Random Forest, SVM, Logistic Regression, Gradient Boosting, or Naïve Bayes can be applied to string statistics, website content, and metadata to distinguish malicious and legitimate URLs [11], [13]. Similarly to phishing detection, researchers also applied deep learning architectures to process raw string data as input. Character-level CNNs are particularly effective at learning local patterns of malicious URLs while LSTMs learn temporal patterns embedded in the URL string [11], [13]. Researchers also utilize Vision Transformers to demonstrate the effectiveness of the self-attention mechanism at parsing long strings of malicious URLs [15].

Most works report relatively high accuracy, F1-score, recall, and precision metrics. However, many solutions fail to consider the limitations of real-time processing and practical deployment scenarios. Moreover, most solutions are evaluated on small-scale benchmark datasets, offering limited insights into their scalability and performance in a production environment. Finally, many researchers do not consider the practical aspects of browser-level URL filtering and provide little information on the interpretability of detection results [12], [15].

C. QR Code-Based Cyber Threat Detection

QR codes are now ubiquitous in day-to-day activities as they enable rapid contactless interactions with a wide range of digital services. However, cybercriminals also attempt to leverage this three-dimensional matrix barcode to deliver various types of malicious content. Attackers can hide phishing links, dangerous application launchers, and even malware inside QR codes. Traditional QR code scanners do not distinguish between harmful and benign content apart from basic reputation checks [16], [17].

Researchers explored various machine learning and computer vision solutions to automatically detect malicious QR content. Most solutions incorporate similar processing pipelines that first parse the QR message and then apply

traditional or deep learning algorithms to detect malicious intent. Computer vision techniques detect QR code orientation, location, and other parameters prior to parsing the matrix content. The parsed character string is then subject to analysis where traditional algorithms such as Random Forest, SVM, or ensemble learning methods attempt to distinguish malicious and benign URLs, domains, or other content [17], [18]. Many works demonstrated that deep learning approaches are effective for parsing QR code content, especially when the matrix is rotated, distorted, or poorly scanned [16].

QR code detection literature has several limitations that hinder the development of practical QR-based attack detection solutions. First, most works only focus on the content of the QR code string neglecting the visual appearance of the matrix. Second, most works do not explore methods to explain the QR code detection results to end-users or provide actionable guidance [16]-[18].

D. Image Manipulation Detection

Modern image editing software democratized the creation of fake images, leading to the proliferation of digital forgeries. Malicious manipulations often utilize digital image editing tools and generative adversarial networks to create convincing visuals. Such images may contain falsified content that can be used to distribute misinformation, create synthetic identities, and deploy various fraud schemes. Earlier research works explored classic digital forensics methods that rely on image metadata, color channel statistics, and typical artifacts introduced by JPEG compression to detect tampering [19], [20]. These methods are relatively easy to implement and demonstrate acceptable accuracy for many practical applications. However, they require manual analysis and are not easily automated. Moreover, these solutions fail to detect sophisticated manipulations that employ modern generative models or involve multiple layers of editing [19], [20].

Researchers are also exploring deep learning approaches that allow for detecting image forgery automatically. Existing literature demonstrates that convolutional neural networks can effectively extract relevant features from both the local region of interest and the entire image. Several works also employ Vision Transformers to learn image-wide patterns. Some recent approaches utilize semantic segmentation to identify the exact region within the image that was modified [21].

The literature on image manipulation detection is limited in several key areas. First, most methods are not reliable when the input image was manipulated using a generation model that was not used during training. Second, many solutions show reduced accuracy when the source images were captured with different compression settings or down-sampling rates. Finally, most solutions are not easily interpretable and do not provide sufficient information on the type or extent of image modification [21].

E. Deepfake Video Detection

Recent advances in video generation models enabled the creation of extremely realistic deepfake media. Deepfake videos are often used to impersonate public figures and spread disinformation, as well as to launch spear-phishing campaigns targeting specific organizations or individuals [22]. Researchers are developing detection solutions that identify signs of forgery by analyzing incongruities between the generated video and real-world physics or human anatomy. Most conventional approaches utilize convolutional neural networks to extract high-level features from video frames, which are then analyzed by various architectures to detect artifacts indicative of deepfake technology [23]. Authors also employ recurrent networks to harness temporal information across successive video frames [23].

Researchers often combine architectures to extract spatio-temporal features to maximize the detection performance. Hybrid CNN-RNN approaches, 3D-CNNs, and Vision Transformers are particularly effective at detecting inconsistencies across consecutive video frames [23]. Various works highlight that researchers also analyze micro-expressions, eye blinking, and larynx motion to identify inconsistencies introduced by deepfake generation software.

Detection of deepfake videos remains a significant challenge since the field evolves rapidly. A deepfake video detection solution that displays excellent results on publicly available benchmark datasets often shows significantly worse performance when confronted with a novel method of video generation. Moreover, typical compression artifacts often mask subtle inconsistencies across video frames [24]. Many works also highlight that current approaches have limited applicability to unconstrained real-world videos and suffer from insufficient interpretability [24].

F. Explainable Artificial Intelligence (XAI) in Cybersecurity

While complex AI models excel at classification accuracy across cybersecurity applications, the models' final predictions are often completely opaque. This presents a major issue when any enterprise security application flags an email, URL, QR code, image, or video as dangerous and requires a human security analyst to take action. If an architecture simply outputs a binary classification of "danger," without explicitly stating which features triggered the classification, analysts cannot easily verify, audit, or trust the model's prediction [25].

Explainable Artificial Intelligence (XAI) addresses this concern by explicitly defining how model predictions are made. Post-hoc interpreters such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), feature importance, self-attention, and other explainers have been used to great extents to explicitly define the inner-workings of any automated cybersecurity application, [26].

Previous literature on intrusion detection, malware classification, and even single-type phishing detection (e.g., only URL) showcases that XAI is critical in enabling human analysts to understand model logic, identify the most influential features, and detect possible false positives. With that said, simply attaching an explanation is not sufficient in all cases: the model's feature attributions must be inherently useful, coherent, and optimized for the task at hand [26].

Perhaps the most significant finding from the literature is that XAI is still heavily dominated by single-domain detection architectures that do not communicate or reason about data from other sources. In other words, phishing, URL, QR code, image, and deepfake models all utilize independent, non-integrated architectures that produce disjointed, unrelated explanations. Little research has explicitly addressed the need for multi-domain explainability, despite the clear value in unifying phishing, URLs, QR codes, and images into a single enterprise cybersecurity application. This is the primary motivating factor behind the CyberShield AI architecture, [26].

TABLE I. COMPARATIVE ANALYSIS OF EXISTING APPROACHES

Domain	**Existing Methods**	**Algorithms**	**Advantages**	**Limitations**
Email	NLP + ML/DL	BERT, LSTM, RF	High accuracy	Limited zero-day detection
URL	Feature-based + DL	CNN, LSTM, XGBoost	Detects patterns	Poor explainability
QR	Image + URL analysis	CNN, ML classifiers	Detects malicious QR	Limited contextual analysis
Image	Digital Forensics + CNN	ResNet, ViT	Detects manipulation	Dataset dependency
Video	Deepfake Detection	CNN-LSTM, Transformers	Detects synthetic media	Poor generalization
XAI	Explainable Models	SHAP, LIME	Improves trust	Limited multi-domain integration

III RESEARCH GAP

The comprehensive review of the existing literature highlights that AI-based cybersecurity solutions demonstrate considerable progression toward the identification of diverse cyber threats. Research works cover the utilization of Machine Learning, Deep Learning, Natural Language Processing, and Computer Vision solutions for phishing detection, malicious URL identification, malware recognition, deepfake spotting, and image authentication [1]–[5]. However, the majority of the considered techniques tackle singular aspects of the problem and are not designed to operate in unison toward the identification of diverse attack vectors.

An inherent limitation of the surveyed research works is the distinct lack of cybersecurity solutions that operate across diverse domains. Contemporary phishing campaigns are increasingly utilizing multi-channel cyber attacks that comprise phishing emails, malicious URLs, QR-code based attacks, and fake image spam to conduct highly sophisticated social engineering schemes. Solutions that operate independently on singular fronts are not sufficient to provide a complete cyber threat intelligence overview, [7].

Another significant limitation of the existing research works is the absence of hybrid risk assessment mechanisms. The majority of AI-based threat identification models operate independently and produce isolated outputs in the form of binary classifications (e.g., malicious/benign or fake/real). However, enterprise-level cybersecurity systems require an integrated assessment of diverse threat indicators to produce a unified risk score and guide the incident response team in determining the most pressing issues to handle first. A hybrid risk assessment paradigm could enable cybersecurity analysts to utilize the produced risk indicators across diverse attack surfaces to arrive at a holistic assessment of the breach.

The existing literature does not sufficiently address the issue of explainability and transparency of the produced AI-based cybersecurity models. While deep learning architectures are known to provide state-of-the-art results, their black-box nature reduces the level of trust and transparency, thus limiting their applicability in practice. While the Explainable AI paradigm provides higher-level insights into the decision-making process of AI models, the incorporation of such model analysis components at the infrastructure level is still limited.

The reviewed research works do not sufficiently utilize the concept of algorithmic adaptability and applicability in zero-day scenarios. The majority of the works are based on the utilization of benchmark datasets that contain a fixed selection of attack vectors, which limits the capacity of such

models to a narrow set of use cases. Cybersecurity threats are continuously evolving as attackers seek to utilize new types of phishing templates, domains, and image generators to circumvent the currently employed AI-based threat detection solutions. An adaptive cybersecurity AI model could be beneficial in terms of offering higher flexibility and improved threat detection rates in the presence of rapidly evolving attack vectors.

The reviewed research works do not offer a comprehensive approach to multimodal threat analysis. Cybersecurity threats increasingly comprise diverse forms of attack vectors that could be utilized in unison or at different stages of an attack. However, the majority of the approaches only tackle isolated aspects of the problem, which limits their ability to recognize interconnections and patterns that could be indicative of a larger cybersecurity threat. The use of multimodal AI-based threat detection systems could be beneficial in terms of establishing connections between diverse attack vectors.

Based on the conducted review of the existing literature, the primary areas of concern in terms of research gaps and improvement opportunities in AI-based cybersecurity solutions could be summarized in the following points:

• Unified Cybersecurity Threat Detection: The majority of the reviewed literature focuses on isolated aspects of the problem. Solutions that utilize a common framework and operate on a combination of attack surfaces could provide a more comprehensive overview of an ongoing cybersecurity threat scenario.

• Hybrid Threat Assessment: The existing solutions operate on isolated principles and produce binary classification results. The use of hybrid risk assessment in which the model combines the risk scores from different domains could assist in the creation of a more comprehensive overview of an ongoing threat.

• Cross-Domain Cybersecurity Analysis: Current research works operate on singular attack surfaces that are not integrated with one another. Such an approach is not conducive to the identification of advanced cyber threats that combine multiple vectors in an effort to obfuscate the attackers' presence and intent.

• Explainability and Transparency: While AI-based cybersecurity solutions demonstrate high levels of accuracy, the current literature does not address the issue of model transparency and explainability. This limits the ability of analysts to independently verify the outputs of AI-based models at the enterprise-level.

• Adaptability and Zero-day Threat Detection: The majority of the reviewed literature is based on the use of AI models that are trained on large-scale benchmark datasets. This restricts the potential applications of such models in the context of zero-day scenarios, where the attack vector has not yet been identified.

• Real-time Enterprise Readiness: The majority of the reviewed research works provide a conceptual overview of the problem and do not fully address the potential of their solutions in real-time enterprise scenarios. The current literature does not fully consider the limitations of existing infrastructural capabilities.

Table II Comparative Analysis Table

Threat Domain	Representative Study	Techniques Used	Major Findings	Research Gap Identified
Email Phishing	AI-Based Phishing with BERT	BERT, XGBoost, NLP	High contextual text classification accuracy.	Weak against zero-day, multilingual, and GenAI emails; lacks model XAI.
Malicious URL	ML-Based Malicious URL Framework	Random Forest, XGBoost, Lexical	Effective lexical and domain feature identification.	Fails on dynamic, shortened, or newly registered domains; poor adaptability.
QR Code Threat	Intelligent QR Code Security System	QR Decode, ML, Reputation	Decodes matrix parameters and checks domain reputation.	Misses visual QR tampering, social engineering context, and active risk scores.
Image Forgery	DL-Based Image Forgery Detection	CNN, ELA, Metadata Analysis	Localized pixel-level and forensic noise analysis.	Vulnerable to hyper-realistic GenAI imagery and deep editing methods.
Deepfake Video	ViT-Based Deepfake Framework	ViT, CNN, Temporal Extraction	Spatio-temporal and multi-frame anomaly tracking.	High performance drop under heavy video compression or unseen GAN models.

VI PROPOSED CONCEPTUAL FRAMEWORK FOR FUTURE RESEARCH

The collective examination of the state-of-the-art literature reveals that the majority of existing AI cybersecurity solutions are focused on addressing singular threat surfaces, which may lead to compartmentalized detection capabilities and limited overall protection for enterprises facing multi-surface attacks. To this end, this section proposes the conceptual reference architecture of CyberShield AI as an

end-to-end conceptual reference framework for multi-domain cybersecurity threat detection and mitigation.

A. Data Acquisition and Structural Preprocessing Layer

The reference architecture begins with the Data Acquisition Layer, which is designed to ingest various formats of unstructured input from enterprise users. This layer is capable of receiving arrays of textual characters (e.g., email content), strings of characters (e.g., URLs), matrices of pixel-values (e.g., QR codes), static visual representations (e.g., digital images), and video files. This layer houses format-specific normalizers that tokenize, normalize, and extract metadata from the input streams for subsequent processing.

B. Threat Detection Layer

The Threat Detection Layer implements AI threat detection modules for classifying phishing emails, detecting malicious URLs, scanning QR codes for embedded threats, examining digital images for manipulations, and video deepfakes. For each module, appropriate state-of-the-art techniques reported in the literature are considered. For instance, transformer-based natural language processing models, ensemble learning methods, and deep learning computer vision techniques have shown potential for different aspects of threat detection [3], [4]. Rather than existing as isolated components, the five threat detection modules operate in parallel to extract domain-specific threat indicators from the input communication asset.

C. Explainable Artificial Intelligence (XAI) Layer

To improve the model's explainability and interpretability, the conceptual architecture includes the Explainable Artificial Intelligence (XAI) Layer, which implements Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) algorithms to generate model-agnostic interpretations of the model's decisions. The literature review phase identified that while deep learning cybersecurity models tend to demonstrate high levels of performance, their implementation often raises concerns about model transparency, verifiability, and explainability, which may affect enterprise adoption rates [11]. In CyberShield AI, the XAI Layer serves as the bridge between the abovementioned AI threat detection modules and enterprise users, transforming the numeric tensors of the model into human-readable explanations. In other words, this layer enables the model to justify its decisions by identifying the tokens, characters, or pixels that contributed to the final prediction regarding the presence of a cybersecurity threat.

D. Hybrid Risk Assessment Layer

The Hybrid Risk Assessment Layer is another critical component of the CyberShield AI reference architecture, as it is designed to consolidate the output of individual threat detection modules and generate a consolidated cybersecurity risk score for the analyzed communication asset. Modern cybersecurity threats often include combinations of phishing attempts, malicious URLs, social engineering, and multimedia-based attacks; therefore, a hybrid approach that utilizes multiple threat detection engines can yield better results compared to the performance of any single module [12]. This layer employs risk-based cybersecurity literature to create a risk matrix that would allow the model to rank the input communication asset based on the produced risk scores.

E. LLM-Based Cybersecurity Assistant Layer

Finally, the framework includes the LLM-Based Cybersecurity Assistant Layer, which aims to utilize large language models to transform the model's predictions into user-friendly natural language explanations and recommend appropriate mitigation actions. The rapid development of the large language models (LLMs) domain enables their potential application in the cybersecurity domain, specifically in threat explanation and mitigation recommendation. Nevertheless, further research is required to consolidate their applicability, reliability, and privacy properties within the cybersecurity domain. This layer essentially acts as an interface that translates the feature importance weights and hybrid risk assessment scores into end-user-friendly language and actionable items.

F. Architectural Synthesis

In summary, the reference architecture presented in this section does not claim to be an ultimate solution to the problem at hand. Instead, it is a conceptual synthesis of the most promising aspects reported in the literature review to demonstrate the potential of multimodal threat detection, explainability, risk assessment, and user-centric interfaces. These four components can be further expanded or reconfigured as new solutions emerge to advance the field of AI-enhanced cybersecurity.

G.Structural Novelty and Comparative Framework Taxonomy

To establish the distinct architectural contributions of *CyberShield AI*, fig.II contrasts it against contemporary baseline paradigms across critical structural dimensions.

Dimension	SOAR Platforms	Multimodal Pipelines	Threat Intelligence Hubs	Proposed CyberShield AI Framework
Modalities	Log telemetry, alerts.	Isolated dual modalities (e.g., Image+Text).	Indicator feeds, static hashes.	**Unified Streams (Text, URL, QR matrices, Image, Video).**
Modeling	Rule-based API tool routing.	Early/late vector fusion.	Simple dictionary matching.	**Concurrent Processing with Cross-Surface Context Variables.**
Data Fusion	Post-alert playbook trigger.	Concatenation or decision voting.	Metadata intersection rules.	**Multi-Stage Hybrid Risk-Weighted Fusion Matrix.**
Correlation	High-level timestamp logging.	Non-existent; single-target focus.	Database relational tracking.	**Active Graph-Theoretic Context Linkage Engine.**

Correlation	High-level timestamp logging.	Non-existent; single-target focus.	Database relational tracking.	**Active Graph-Theoretic Context Linkage Engine.**
Explainability	Plain dashboard metrics.	Isolated post-hoc SHAP/LIME.	None provided.	**Unified Cross-Domain Feature Maps via Generative NLP.**
Risk Model	Static CVSS weighting.	Softmax baseline probabilities.	Reputation indices/blocklists.	**Dynamic Correlation-Weighted Risk Scoring Profile.**
Mitigation	Automated endpoint scripts.	Hard binary threat alerts.	Manual policy distribution.	**Interactive Generative Logs with Remediation Guides.**
Validation	Production SIEM environments.	Lab benchmark databases.	Global native repositories.	**Simulated Multi-Stage Cross-Surface Scenarios.**

Fig II Comparative Framework Taxonomy and Novelty Mapping

As detailed in fig II, the novelty of CyberShield AI is not a combination of individual elements. It involves the invention of the Cross-Modal Artifact Correlation Engine – a novel way to correlate dissimilar features (e.g., phishing text patterns and image compression artifacts) in a space defined by shared graph theory principles. This reveals otherwise inaccessible mathematical relationships between attack phases that cannot be captured by disjointed unimodal approaches.

The approach also invents a novel cross-domain representation that effectively replaces conflicting post hoc SHAP and LIME values for different modalities with a set of spatial and textual attribute weights. These weights can be used to train an embedded language model to produce a coherent, explainable, and human-interpretable sequence of operational security recommendations.

H. LLM Security, Privacy, and Governance Framework

The LLM Assistant Layer implements the following security and governance mechanisms to prevent injection attacks, prevent privacy leaks, and ensure the model adheres to the organization's policies:

• The system employs "prompt injection isolation" by utilizing Structured, Bounded Inputs. This is achieved by sanitizing inputs/output from any raw executive control functions. For instance, if the system receives an email message or QR code as an input, it will sanitize the input to remove any raw executive control functions;

• The system utilizes "policy grounding" to prevent the model from providing dangerous or inappropriate recommendations. It is achieved by applying Retrieval-Augmented Generation based on a local vector database that contains information on appropriate and secure recommendations;

• The system introduces a "human in the loop" recommendation approval gateway. This feature is designed to ensure that all recommendations are reviewed by a human representative who approves the input and output of the model before any recommendation is given;

• The LLM Assistant Layer is equipped with the tools to ensure privacy and auditing capabilities. The layer utilizes zero-retention data storage and session duration to ensure that data is not stored in the system for longer than necessary. Additionally, the model's outputs contain audit logging and confidence scoring to indicate the likelihood that the model's output is correct.

VII. DISCUSSION AND RESEARCH IMPLICATIONS

Analysis of the state-of-the-art intrusion detection approaches demonstrates that the majority of them are already quite effective in addressing specific aspects of the cybersecurity landscape. Techniques from the fields of Machine Learning, Deep Learning, Natural Language Processing, and Computer Vision have demonstrated tremendous success in detecting phishing emails, suspicious links, anomalous program behavior, image manipulation, and deepfake video spam–. However, most of these approaches work in isolation, targeting only one particular class of cyber threats. Such solutions lack the ability to combine the evidence from different sources and, thus, miss the wider context of the unfolding attack. The rise of multi-platform cybercriminal operations that simultaneously exploit various classes of network data produces a necessity to rethink traditional intrusion detection approaches. Modern cybersecurity threats are no longer confined to one particular method – threat actors employ a battery of common techniques to maximize the effect of the attack. For example, phishing campaigns can include not only spam emails but also specially crafted QR codes, suspicious URLs, and video attachments. While each element presents a unique threat, relying on isolated detection tools makes it impossible to observe the big picture. A number of recent works stress the importance of consolidated security solutions that embrace the ability to combine different intrusion detection models to produce an integrated analysis of the ongoing attack.,

In this regard, one can note that the cybersecurity field has an urgent need to develop multimodal intelligence tools. Current artificial intelligence (AI)-powered approaches demonstrate outstanding performance but are typically designed to process one particular data type. The separation between text, numerical data, image, and video forms a serious obstacle for building an end-to-end system that could utilize such information concurrently. However, converting these data types into a single representation space may allow researchers to obtain a clearer perspective on the attack surface. Several recent works confirm the benefit of multimodal analysis for intrusion detection,, which provides sufficient motivation for further exploration in this direction.

The surveyed article also sheds light on the importance of the necessity to incorporate Explainable AI (XAI) concepts into cybersecurity applications. While deep learning models

demonstrate excellent accuracy, their internal workings remain unclear, which complicates their immediate application in practice. Cybersecurity analysts require proper justification of the model's decisions before incorporating its suggestions into their operations. Fortunately, recent advances in the fields of AI and cybersecurity allow one to implement post-hoc explainability methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations). These techniques help elucidate the manner in which an AI model arrived at a specific conclusion, thus, allowing cybersecurity professionals to verify the result before taking any concrete action.,

At the same time, cybersecurity models need to evolve further to accommodate the rapid growth of adversarial examples. Cybercriminals continue to develop new methods to obfuscate their operations, which makes it difficult to train AI models capable of withstanding such attacks. Most of the current advanced intrusion detection systems are based on static examples, which limits their ability to detect novel attack vectors. Researchers should devote more attention to the implementation of innovative machine learning techniques capable of detecting polymorphic malware, adversarial phishing emails, and similar advanced persistent threats. Exploring the benefits of integrating federated learning and threat intelligence systems may prove beneficial for advancing the current cybersecurity paradigm,

One of the emerging trends that could benefit cybersecurity is the appearance of Large Language Models (LLMs). Some researchers utilize LLMs in various tasks, including security-related operations. For example, such models can be helpful in retrieving relevant information from the scientific literature, analyzing attack-specific details, explaining the nuances of code injections to developers, and providing general guidance on the threat landscape. Essentially, LLMs can support cybersecurity analysts in their day-to-day operations, providing them with clear explanations of the observed anomalies. However, prior to implementing such models in practice, one has to overcome several challenges, including hallucinations, adversarial prompting, privacy preservation, and general applicability.

Each of the mentioned directions is of paramount interest for the cybersecurity field. The article under review attempts to provide a glimpse of the general idea behind each of these projects. CyberShield AI is an interesting concept that provides a rough vision of how different areas of AI can be integrated into a single software solution. While individual detectors provide higher accuracy, their combination into a unified pipeline allows for building a more comprehensive perspective on the threat landscape and utilizing multiple data sources concurrently. The described methods can be implemented separately or used in conjunction with each other to create a novel approach to cybersecurity.

X. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

While AI-based cybersecurity tools demonstrate extraordinary potential for detecting various digital threats, certain roadblocks need to be overcome before full-fledged and integrated real-time defensive applications can be developed. While the CyberShield AI reference architecture presents an initial set of design principles to address certain shortcomings, its practical implementation will involve a number of significant technical and security-related challenges [1]–[5].

A. Main Challenges of Research

• Insufficient Quality of Multimodal Benchmarks: A critical limitation in the current research space is the scarcity of high-quality multimodal cybersecurity benchmarks. While most of the currently available training sets are dedicated to a specific domain, addressing the unique patterns of phishing texts, malicious URLs, or spatial deepfakes, an integrated system will require a large-scale and diverse benchmark that will also include text-image associations (QR codes), network pathways, and multimedia content. The absence of such a dataset will make it exceptionally challenging to train the algorithm to detect complex associations between various features, [2].

• Need for Real-Time Processing and High Computational Costs: The necessity to execute multiple inference threads in parallel will impose extraordinary requirements on the system's memory and processing power. Performing simultaneous operations with feature extractors, post-hoc explainers, and hybrid risk assemblers will result in significant delays and computational overhead. Such operations will need to be optimized extensively to enable real-world applications in enterprise environments. Future research will involve exploring ways to implement model compression and utilize domain-specific compression algorithms that will help reduce the overall overhead, incurred by running multiple threads simultaneously.

• Generalization Across Various Attack Vectors: Cybercriminals will continue to update and re-label their data to obfuscate their intentions and increase the likelihood of reaching their target. As a result, the performance of many AI-powered models will drop considerably because most algorithms have difficulty generalizing across various attack surfaces. Future research should explore the concept of generalized zero-day attack detection and consider ways to adjust the weights of an already-trained model in real-time to adapt to the changing input data.

• Need to Preserve Privacy and Adopt Secure Infrastructure: Most practical applications of AI-based security systems involve receiving and analyzing highly-sensitive private user data, including emails, payment operations, images, and network traffic. Using such datasets introduces significant security risks and raises major privacy concerns, particularly in the context of increasingly stringent data protection regulations. To address these issues, future research should explore the concept of privacy-preserving machine learning, including developing algorithms that utilize federated learning methods in distributed networks to analyze data without direct access to it.

• Multi-Modal Data Interpretation and Reliability: While post-hoc explainers can provide reasonable justification for isolated model decisions, generating reliable and consistent results with multimodal data remains an issue. Future research should focus on developing a comprehensive XAI framework that would be able to integrate and visualize the decisions made by isolated modalities and identify the associations between them.

• LLM and Agent Vulnerabilities: The use of Large Language Models and autonomous agents as interactive assistants is an innovative concept, as it allows providing end-users with essential information about the detected threats. However, current implementations remain vulnerable to attacks and present certain security and reliability risks due to the potential for hallucination and the challenges of introducing changes. Future research should focus on eliminating these vulnerabilities to ensure that such tools can be used in production-grade environments.

• Interacting Tools for Human Analysts: Most AI-powered security applications pursue an end-to-end approach and attempt to optimize the entire decision-making pipeline. However, such systems remain highly susceptible to errors and frequently require manual review by human analysts. Instead, future research should include the creation of an interactive pipeline that will support collaboration between AI and human security analysts.

B. Research Roadmap

To ensure that future research efforts are aligned with the current requirements, the essential components of the CyberShield AI design could be prioritized as follows:

1. Domain-Specific Multimodal Benchmarks: Creating a high-quality open-source dataset that will contain numerous multimodal correlated examples of phishing emails, spam texts, malicious URLs, QR codes, and spatial deepfakes, [2].

2. Reducing Resource Costs with Optimized Architectures: Designing an optimized multimodal application that will minimize the resources needed for processing data and support large-scale inference across various modalities.

3. Generalized Zero-Day Attack Detection: Developing an advanced real-time application that will utilize a continual learning pipeline to detect zero-day attacks and update the weights of a pre-trained model.

4. Privacy-Preserving Distributed Learning: Integrating novel cryptographic algorithms to optimize the security of distributed and federated learning and ensure that sensitive data remains protected throughout the process.

5. Unified Post-Hoc XAI Explainers: Developing an integrated model that will support multimodal security analysis and visualization to help security analysts review the model's decisions and ensure that they are not biased.

6. LLM Operator Security: Integrating novel methods and algorithms to secure LLM operators and ensure that end-users receive accurate and unbiased information regarding the potential threats.

C. Final Remarks

In general, the future of cybersecurity research is to develop tools that will be able to detect an extensive array of digital threats. However, there is a growing need to ensure that such systems preserve the privacy of users, remain transparent, and operate in a manner that supports effective collaboration between AI and human security analysts. Overall, integrating various components, including multimodal machine learning, post-hoc explainers, continuous learning, and human-in-the-loop pipelines, will help create a robust cybersecurity framework that will be able to address the complexity of modern threats [4]–[7].

X. CONCLUSION

The global cyber threat landscape evolves at a rapid pace, making legacy signature-based security solutions increasingly obsolete. This development necessitates adaptive systems that can detect, triage, and respond to multifaceted, multipronged cyber offenses. This review article demonstrates that artificial intelligence technologies, including traditional machine learning methods, deep learning networks, natural language processing, computer vision, and explainable AI, have shown promising results in addressing specific and singular security tasks, such as phishing text detection, URL tracking, QR code scanning, image forensics, and deepfake video detection. Still, contemporary research practices indicate that the bulk of the field's progress is focused on studying isolated threats and single-source security analytics. In other words, the prior art lacks holistic and integrative solutions capable of withstanding and analyzing ongoing multipronged cyberattacks.

As such, this review analyzes the current state-of-the-art solutions from the perspective of the cybersecurity landscape of 2024 and identifies a set of challenges and areas of interest. The themes include the absence of an inter-domain framework, the lack of multimodal security analysis, and the need for hybrid risk assessment methodologies. At the same time, there is a clear need for improved model interpretability and the ability to adapt the existing solutions to the constantly evolving threat landscape. This state of affairs informs the conceptual basis and design choices of CyberShield AI, an envisioned modular framework that would consolidate the efforts and resources of an organization by unifying multiple security-specific AI/ML disciplines under a single umbrella.

CyberShield AI is designed to facilitate the exchange of information between submodules that analyze phishing texts, sequential URLs, QR codes, still images, and deepfake videos. Thus, this framework enables a high-level conceptual understanding of ongoing cyberattacks and emergent security threats. CyberShield AI emphasizes bottom-up information exchange and post hoc explainability throughout the pipeline, as well as a risk assessment methodology to consolidate the output of different submodules into a unified action space. Finally, it includes a large language model–based cybersecurity advisor module that operates as an end-user interface, explaining the rationale behind model outputs in natural language.

It should be noted that the review only provides a conceptual basis for such a framework and does not elaborate on the technical design decisions. At the same time, CyberShield AI can serve as a theoretical foundation for future research in the field. There is a need to advance the current theoretical understanding, develop practical applications, and identify new frontiers. The technical literature on the application of AI to cybersecurity is highly prolific in this regard, calling for further system-level innovations. Future studies should consider the creation of a unified, horizontally scalable cybersecurity operations framework that will consolidate the vertical security solutions of the past decade.

Such an endeavor would require extensive collaboration between researchers and practitioners to collect the required data sets, optimize the solutions for production environments, and minimize the resource footprint of AI-powered security applications. Specifically, the process would involve reducing the power requirements and time-to-interpret parameters of new algorithms in horizontal cybersecurity operations. At the same time, novel solutions should include built-in privacy-preserving mechanisms, horizontal model adaptation techniques, continuous pretraining pipelines, and specialized defenses against large language model (LLM) attacks. In sum, the article highlights the importance of transitioning from the existing approaches to isolated AI-assisted security solutions in favor of a more transparent, unified, and holistic approach to cybersecurity.

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